

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

Neural Network Models for Extracting User Attributes from Dialogues

by

Dang Ngoc Son

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Supervisor:

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Abstract

In this thesis we present an approach to extracting user attributes that can be represented as a *subject-predicate-object* (SPO) triplets from utterances in dialogues. We view user attribute extraction as a pipeline of two tasks: predicate classification and entity generation. The predicate classification task is to determine predicates triggered in a sentence and the entity generation task is to generate corresponding subjects and objects for each triggered predicate. Since multiple predicates may be triggered in one sentence, we formalize predicate classification as a multi-label classification problem and fine-tuned a pre-trained BERT model to solve the problem. We formalize the entity generation task as a sequence-to-sequence problem and fine-tuned T5 model on the task data to create the entity generation model. In the best of our knowledge, our work is the first work of using T5 model for the entity generation task. Experimental results show that the proposed deep learning model outperforms a state-of-the-art baseline model by a large margin.

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Chapter 1 Introduction

1 Introduction

Intelligent assistants have become an essential part of our life, being able to deal with a wide range of routine tasks and provide factual information. There is still a growing demand for creating user-oriented chatbot agents, which are able to hold personalized conversations. For meaningful and diverse dialogues with a real person, a system should be able to infer knowledge about the person's background from user utterances. Such rich information reveals important clues of user information including biographic facts (e.g., age, gender, relationship), personal interest (e.g., politics, technology, entertainment, sports), occupation information (e.g., researcher, student, software engineer, musicians), emotional orientation (e.g., optimistic, negative), etc. We call these personal information *user attributes*. Inferring such user attributes can benefit many applications in user profiling, information retrieval, personalization, and recommendation. They provide a rich repository of personal information for better user understanding in many applications. High-quality user attributes are, however, hard to obtain from utterances in a dialog. Therefore, exploiting unstructured data sources to obtain structured user attributes is a challenging research direction. Most existing work has treated user attributes separately and studied user attribute prediction independently, or only extract a few types of user attributes such as gender, age, relationship, occupation, interest, and emotional orientation. We argue that the user attribute relation is an important characteristic and will facilitate accurate user attribute inference. Moreover, exploration of attribute relations will enable novel mining and application scenarios. This serves as the basic premise of this thesis. In this thesis, we demonstrate that it is feasible to automatically extract user attributes from dialogues. Given a user utterance, our goal is to predict user information that can be represented as a (Subject, Predicate, Object) triplet format, which is available for any downstream application. Consider the following conversation between a person (P) and a system (S):

S: *Hello, how are you doing today?*

P: *I would like to go traveling!*

S: *You went to London the other day, didn't you?*

P: *I just visited London, which was a disaster. My hotel was a headache, and I spent half the time in bed with a fever...*

For the above example, (*I, like_activity, traveling*) is extracted from the second user utterance. Meanwhile, not every utterance has useful information, and some have multiple attributes. For instance, “*How are you doing today?*” does not have any user-specific information. In this study, we view user attribute extraction as a pipeline of two tasks: the predicate prediction and entity generation. The predicate prediction task first determines whether there is a predicate triggered by a user utterance. This is considered as a multi-label classification problem because there could be zero or multiple attributes. If there is a triggered predicate, then the entity generation task further generates the subject and object phrases to complete the whole user attribute. In this thesis, we show that we can significantly improve the accuracy of user-attribute extraction by using pre-trained language models for both tasks. Additionally, to solve the imbalanced classes in the predicate prediction task, we propose two methods that are Class-Weight and Data Enrichment approaches. We empirically show that our strategy outperforms several retrievals and generation baselines.

Our contributions are summarized as follows:

- We propose an attribute extractor, train our model on a Distant Supervision Dataset, leverage transfer learning with a pre-trained language model, and apply Class-Weight to obtain effective training samples.
- We leverage transfer learning with pre-trained language models for both predicate classification and entity generation steps. In the predicate-classification step, we used class-weight and data enrichment in training to reduce the data imbalance problem.
- We are the first of using the T5 pre-trained language model for entity generation.
- We significantly improve a state-of-the-art baseline in the user-attribute extraction on public data.
- We discuss potential applications of the extracted user attributes, point out current limitations, and future works.

Our thesis is organized into seven chapters. In Chapter 1, we introduce the motivation, purpose of our research, and main contributions. In Chapter 2, we present research problems in the user

attribute extraction task, the existing processing methods, and problem formalization. Chapter 3 describes transfer learning with pre-trained language models. Our proposed methods for two tasks are presented in Chapter 4 and Chapter 5, respectively. We describe experiments and results in Chapter 6, and finally, we give the conclusions in Chapter 7.

Chapter 2

User Attribute Extraction from Dialogues

In this chapter, we introduce the task of user attribute extraction from dialogues and some existing methods to address this task. This chapter aims to provide audiences with basic knowledge related to user attribute extraction from conversations.

2 User Attribute Extraction from Dialogues

2.1 User Attribute Extraction

Extracting user attributes is an important task in a Personal Artificial Intelligence (P.A.I) system to acquire information and knowledge through conversations between the system and humans. Inferring such user attributes can benefit many applications in user profiling, information retrieval, personalization, and recommendation. They provide a rich repository of personal information for better user understanding in many applications. High-quality user attributes are, however, hard to obtain. Therefore, exploiting unstructured data sources to obtain structured user attributes is a challenging research direction. Let's consider the user's utterance: *"My son. I bring him to church every Sunday with my Ford."*. As humans, we can infer attributes $\langle\langle I, have children, son \rangle\rangle$, $\langle\langle I, like activity, church \rangle\rangle$, $\langle\langle I, have vehicle, ford \rangle\rangle$. Capturing such user traits is a daunting task, however, with both implicit and explicit signals presented. We need to consider the context of user's utterance, to infer user's attribute. In this example, cues *"My son"*, *"my ford"*, *"church"* are necessary contextual words to predict the attributes *"like activity"*, *"have vehicle"*, *"have children"*.

In this study, we view the user attribute extraction problem as a pipeline of two tasks: predicate prediction and entity generation. The predicate prediction task is to determine predicates triggered in a user utterance. This is considered as a multi-label classification problem because there could be zero or multiple attributes in a user utterance. If there is a triggered predicate, then the entity generation task further generates the subject and object phrases to complete the whole user attribute.

2.2 Datasets

Persona-Chat [1] This is a multi-turn chit-chat corpus with annotation of the participants’ personal profiles (e.g., preferences about food, movies). It is collected by asking two crowd-workers to talk to each other freely but conditioned on their artificial personas, which are established by four to six persona sentences. The Persona-Chat dataset contains multi-turn dialogues conditioned on personas. The dataset consists of 8939 complete dialogues for training, 1000 for validation, and 968 for testing. Each dialogue was performed between two crowd-sourced workers assuming artificial personas (described by 3 to 5 profile sentences, such as “*I like to ski*”, “*I am an artist*”, “*I eat sardines for breakfast daily*”). There are 955 possible personas for training, 100 for validation, and 100 for testing. Additionally, a version of revised persona descriptions is also provided by rephrasing, generalizing, or specializing the original ones. In order to extract the user attributes, they automatically labeled personas with profession and gender attributes by looking in the persona description for patterns “*I am/I’m a(n) <term>*”, where *<term>* is either a profession label or a gender indicating noun (*woman, uncle, mother, etc.*). They manually labeled persons with family status by identifying persona descriptions containing related words (*single, married, lover, etc.*) and labeling the corresponding persona as single or not single.

ConvAI2 The ConvAI2 NeurIPS competition aimed at finding approaches to creating high-quality dialogue agents capable of meaningful open-domain conversation. The ConvAI2 dataset for training models is based on the Persona-Chat dataset. The speaker pairs each have assigned profiles coming from a set of 1155 possible personas (at training time), each consisting of at least 5 profile sentences, setting aside 100 never seen before personas for validation. As the original Persona-Chat test set was released, a new hidden test set consisted of 100 new personas, and over 1,015 dialogs were created by crowdsourced workers. To avoid modeling that takes advantage of trivial word overlap, additional rewritten sets of the same train and test personas were crowdsourced, with related sentences that are rephrases, generalizations or specializations, rendering the task much more challenging. For example, “*I just got my nails done*” is revised as “*I love to pamper myself on a regular basis*” and “*I am on a diet now*” is revised as “*I need to lose weight.*” The training, validation and hidden test sets consist of 17,878, 1,000, and 1,015 dialogues, respectively.

Dialogue NLI [3] This is a new dataset built upon Persona-Chat, which provides a corpus for NLI tasks in dialogues. The authors demonstrate that consistency of dialogue agents can be improved by re-ranking responses using an NLI model. Dialogue NLI consists of sentence pairs labeled as entailment, neutral, or contradiction. The authors first required human annotation of all the persona sentences in Persona-Chat, mapping sentences into triplets $(e1, r, e2)$, where $e1$ and $e2$ are entities and r is the relation type. They pre-defined around 60 different relation types such as “*live in general*”, “*like food*”, and “*dislike*”. Then they group different persona sentences with the same triplet together.

2.3 Problem Formalization

We formalize the user attribute extraction task as follows. Let us denote N utterances in a dialogue as $U = \{u_1, \dots, u_N\}$, where odd and even turns are represented as user utterances and system responses, respectively. M natural language persona sentences $P = \{p_1, \dots, p_M\}$ in the dataset have their corresponding triplets $T = \{t_1, \dots, t_M\}$. Besides persona sentences, each of the utterances may have zero, one, or multiple triplets selected from T . We design two steps to obtain SPO (*subject, predicate, object*) triples from dialogues 1) Predicate Classification 2) Entity Generation.

Predicate Classification: The goal of predicate classification is to assign a set of relevant predicates for a single u_i utterance. Consider the user utterance: “*i got a job working in advertising last year.*”, we would like to identify the predicate triggered “*has_profession*” in the utterance. However, one sample can belong to more than one predicate. Let see another sample: “*sure, i pick him up for church every Sunday with my ford pickup*”, we can identify predicates “*have_vehicle*”, “*like_goto*” in the user's utterance. In order to do that, we cast the predicate classification task as a multi-label classification problem. Let define Pr predicates, $Pr = \{P_1, P_2, \dots, P_N\}$. The aim in multi-label text classification is to assign a subset of Pr to a given user's utterance u_i .

Entity Generation: Given an utterance u_i and a predicate P_i determined by the predicate classifier, the entity generator is to identify a subject (S) and an object (O) corresponding to the P_i . For instance, give the utterance “*i would rather eat chocolate cake during this season.*” and predicate “*like_food*” triggered in the sentence, we would like to extract subject “*i*” and object “*chocolate*”. In order to extract subject (S) and object (O) from predicate triggered (P) and user's utterance (U), we formalize the task as the sequence-to-sequence (Seq2Seq) model. The user's utterance u_i , predicate triggered P will be considered as input sentence as format “ $P ; <s> u_i </s>$ ”, and the output sentence is a sequence “ $S ; O$ ”. To let the model know which task is needed to be resolved, a prefix text “*summarize:*” is needed. For example, the input sequence is “*summarize: live_in_general ; <s> my family lives down the street from me. </s>*”, and corresponding output sequence “*my family ; down the street from me*” will be used for training.

2.4 Challenges

User attributes extraction is an important and well-studied NLP task to acquire information and knowledge. However, extracting a large number of user attributes and their respective values make this a challenging task. We find some difficulties as below. Firstly, generating subject (S) and object (O) is difficult because the same meaning can have many different expressions. Therefore, a large enough amount of training data is needed to cover all cases. So, most importantly, a suitable dialogue dataset with clean attribute extraction labels is needed. Secondly, there is an ambiguous attribute inference problem. This problem suggests that using the model may not always capture the real causality information. For example, the fact that a person attends many parties does not necessarily mean they like parties. Moreover, the textual cues in user conversations are often implicit and thus difficult to learn. For example, consider the user's utterance: “*jeb and george are the names of my 2 siberian huskies*”. As humans, we can infer the user attribute triplet $\langle\langle i, have, 2 dogs \rangle\rangle$. Capturing such user traits is a daunting task, however, with both implicit and explicit signals present. We need to consider the context to infer that attribute. Proper inference requires both deep language understanding and background knowledge (e.g., *jobs, hobbies, cities, etc.*). Lastly, the training data for the entity generation part is also quite large, so the model training time is long.

2.5 Existing Methods

Learning Personal Attributes from Conversations [5] Hidden Attribute Models (HAMs) for inferring personal attributes from conversations, such as a person’s profession, age, occupation, or family status, from conversation using deep learning. The underlying neural network learns to predict a scoring of different object (e.g., different professions) for a given subject-predicate pair by using attention within and across utterances to infer object values. For example, that a subject who often uses terms like theory, mathematical, and species is likely to be a scientist, while a subject who use terms like senate, reporters, and president may be a politician. Tiginova et al. demonstrated the viability of their approach in extensive experiments considering several attributes on three datasets with diverse characteristics: Reddit discussions, movie script dialogues and crowdsourced conversations. They compared HAMs against a variety of state-of-the-art baselines, and achieved substantial improvements over all of them, most notably by the $HAM_{CNN-attn}$ and HAM_{2attn} variants. However, in the data, the distribution of age categories is highly imbalanced, with adult characters dominating the dataset (58.7%) and child being the smallest category (1.7%). They do not consider the family status predicate for MovieChAtt, because the necessary information is often not easily available from Wikipedia articles. Similarly, they do not consider the age predicate on the Persona-Chat [1] dataset, because this attribute is not easily found in the persona descriptions. More generally, all users in the datasets are labeled with exactly one attribute value for each predicate. In a real setting it may be impossible to infer any attribute value for some users, whereas other users may have multiple correct values.

Personal Knowledge from Dialogues: Awareness of background knowledge about users is important in both goal-oriented dialogues [6, 16] and chitchat settings [20, 43]. Such personal knowledge may range from users’ intents and goals [6, 42] to users’ profiles, with attributes such as home location, gender, age, etc. [16, 20, 22], as well as users’ preferences [25, 43]. Prior works on incorporating background knowledge model personas and enhance dialogues by injecting personas into the agent’s response-generating model [20, 22, 43] or by adapting the agent’s speech style [16]. These approaches typically use latent representations of user information. In contrast, there is not much work on capturing explicit knowledge based on user utterances. Li et al. encode implicit speaker-specific information via distributed embeddings that

can be used to cluster users along some traits (e.g., age or country) [20]. Zhang et al. describe a preliminary study on predicting simple speaker profiles from a set of dialogue utterances [43]. The most related prior work by Garera and Yarowsky captures latent biographic attributes via SVM models from conversation transcripts and email communication, taking into account contextual features such as partner effect and sociolinguistic features (e.g., % of “*yeah*” occurrences), in addition to n-grams [12]. Several works explore information extraction from conversational text to build a personal knowledge base of speakers given their utterances, such as extracting profession: software engineer and *employment_history*: Microsoft from “*I work for Microsoft as a software engineer*”, using maximum-entropy classifiers [15] and sequence-tagging CRFs [21]. However, this approach relies on user attributes being explicitly mentioned in the utterances. In contrast, our method can infer attribute values from implicit cues, e.g., “*I write product code in Redmond.*”

User Information Extraction for Personalized Dialogue Systems [7] Toru Hirano et al. proposed a method to extract user information in a structured form, $\langle \textit{predicate-argument structure, entity, attribute category, topic} \rangle$ for personalized dialogue systems. They focus on solving problems in extracting predicate argument structures from question answer pairs in which arguments and predicates are frequently omitted, and in estimating attribute categories related to user behavior that a method using only context words cannot distinguish. It said that the extracted user information should satisfy the following requirements 1) It should have information to reproduce what users said in order to tell users their past utterances. 2) It should have information to complement unknown conditions. 3) It should have information that can be searched to determine which information to provide. 4) It should have information to determine when systems will use which user information.

To satisfy requirement 1, they extract the predicate-argument structure (PAS), which represents “*who did what to whom.*” PAS is useful for representing the basic content of an utterance. Higashinaka et al. (2014) proposed a method to generate system utterances from PAS. For requirements 2 and 3, they extract entities such as person and location names, which represent keywords. For requirement 4, they extract attribute categories such as hobbies and experiences, which represent aspects of users, to determine which information will be used, and we extract topics such as music and travel, which represent main subjects, to determine when to use the

information. However, only for Japanese and its dataset is not public, or the PAS analysis of ellipsis types of proposed methods was not high.

User Attribute Extraction from Dialogues [41] CS Wu et al. proposed a method to extract user attributes which can be represented as a (Subject, Predicate, Object) triplet format, which is available for any downstream application. For example, (*I, live in, Florida*) is extracted from the second user utterance “*I am originally from California but now I live in Florida for long.*”. Meanwhile, not every utterance has useful information, and some have multiple attributes. For instance, “*How are you doing today?*” does not have any user-specific information. They leveraged the state-of-the-art natural language inference (NLI) model to train our model via distant supervision. Using the existing Persona-Chat dataset, comprising dialogues collected given artificial speaker information called personas, they hypothesize that if an utterance is entailed by a persona sentence, then such a persona sentence can be viewed as a valid user attribute. For example, if the persona sentence “*I was a banker*” is entailed by the user utterance “*I no longer work at banks,*” then they can extract the (*I, previous profession, banker*) attribute for the utterance. Although NLI mapping may include some noise, these annotations are cheap and can at least provide a weak source of supervision. A drawback is, there is an ambiguous attribute inference problem. This problem suggests that using the entailment model may not always capture the real causality information. For example, the fact that a person attends many parties does not necessarily mean they like parties. Next, the pre-defined predicates are not collected comprehensively, which may not be able to cover all the relations in a real scenario. Therefore, using clustering techniques to group more predicates automatically is an appealing solution. Lastly, the conversations in the Persona-Chat dataset are not collected naturally, with most of the users tending to ignore what the other said and just talking about themselves. Therefore, the result was not high as their expected.

Chapter 3

Transfer Learning with Pre-trained Language Models

In this chapter, we introduce transfer learning and transformer-based language models, using State-of-the-Art Pre-trained Neural Network Models (OpenAI's GPTs, BERTs, ELMos, ...etc) to Tackle Natural Language Processing Problems with Transfer Learning.

3 Transfer Learning with Pre-trained Language Models

3.1 What is transfer learning

Natural language processing (NLP) has witnessed many substantial advancements in the past three years. With the introduction of the Transformer and self-attention mechanism, language models are now able to learn better representations of the natural language. These attention-based models have achieved exceptional state-of-the-art results on various NLP benchmarks. One of the contributing factors is the growing use of transfer learning.

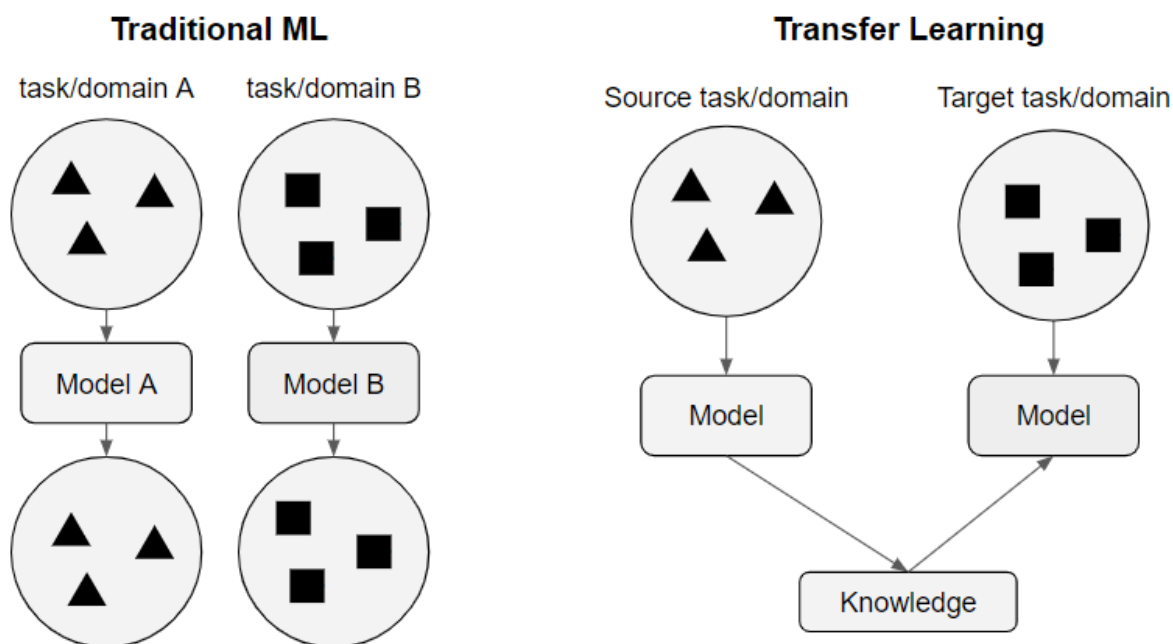


Figure 1: Classic Machine Learning and Transfer Learning

For classical machine learning a model is trained for every special task or domain. Transfer learning allows us to deal with the learning of a task by using the existing labeled data of some related tasks or domains. Tasks are the objective of the model. e.g., the sentiment of a sentence, whereas the domain is where data comes from. e.g., all sentences are selected from Reddit. In the example above, knowledge gained in task A for source domain A is stored and applied to the problem of interest (domain B).

Generally, transfer learning has several advantages over classical machine learning: saving time for model training, mostly better performance, and not a need for a lot of training data in the target domain. It is an especially important topic in NLP problems, as there is a lot of knowledge about many texts, but normally the training data only contains a small piece of it. A classical NLP model captures and learns a variety of linguistic phenomena, such as long-term dependencies and negation, from a large-scale corpus. This knowledge can be transferred to initialize another model to perform well on a specific NLP task, such as sentiment analysis (Malte and Ratadiya 2019).

3.2 Transformer-based Language Models

Transformer-based language models are now widely used in Natural Language Processing (NLP). Transformer-based language models (such as BERT, XLNet, T5, GPT-3) are the most common architecture of language models. Such models can be used to create effective solutions for a broad range of natural language processing tasks, which is why they are widely used in natural language processing.

Jacob Devlin et al. (Oct 2018) [9] proposed BERT, which stands for Bidirectional Representations from Transformers. It marks the beginning of one of the most influential concepts that changed the entire NLP scenario, a pre-trained deep bidirectional model. The term bidirectional emphasizes BERT's context-sensitive word embedding generated by combining the left and right contexts. To train such representation, the model uses a masked language model (MLM), which is inspired by the cloze task [10]. It also utilizes next sentence prediction (NSP), a second pre-training task along with MLM that builds a better understanding of the relationship

between the sentences. To sum up, BERT attains new state-of-the-art results on eleven NLP tasks and establishes a firm foundation for the upcoming language models.

Zhilin Yang et al. (June 2019) [11] proposed XLNet, a generalized autoregressive pretraining method that uses permutation language modeling to enable an autoregressive model to learn bidirectional contexts. The bidirectional nature comes from the permutation of the factorization order, which allows it to see tokens occurring to the left and right in the same sequence. This makes XLNet perform better on language understanding tasks like question answering, reading comprehensions, etc.

Colin Raffel et al (Oct 2019) [4], proposed T5, which stands for Text-to-Text Transfer Transformer. It is the result of combining the best design traits from past work in Language Modelling. The model introduces a unified text-to-text framework where NLP tasks are treated as a text-to-text problem. This novel framework enables T5 to solve diverse downstream tasks using a single pre-trained model, as opposed to fine-tuned task-specific models. T5 achieved the state-of-the-art solution on GLUE, CNN/Daily Mail [14] with achieving near-human score on SuperGLUE.

Alec Radford et al. (June 2020) [23] proposed GPT-3, the third- generation autoregressive language model in the GPT series. The model consists of 175B parameters, the largest in any model till date and is 100x bigger in comparison to GPT-2 [24]. The model implements few-shot learning, i.e., capable of performing a novel NLP task by looking at a limited number of examples/instructions. That is, the model does not update any parameter-weights and can be applied directly on any NLP tasks. In most cases, this eliminates the need to fine-tune the model further, which can be challenging at times due to the unavailability of domain-specific data. GPT3 is task-agnostic and surpasses the previous state-of-the-art fine-tuned model in few downstream tasks.

3.3 Transfer Learning with Pre-trained Language Models

In NLP, transfer learning techniques are mainly based on pre-trained language models, which repurpose and reuse deep learning models trained in high-resource languages and domains. The pre-trained models are then fine-tuned for downstream tasks, often in low-resource settings. The downstream tasks include part-of-speech tagging, text classification, and named-entity recognition, among others. Transformer-based pre-trained language models have proven to be effective for many natural language processing (NLP) tasks, such as text classification and question answering, and they have enabled systems to outperform previous state-of-the-art approaches. A prime example of such language representation models is Bidirectional Encoder Representations from Transformers (BERT), which was pre-trained on the Book Corpus and English Wikipedia (Devlin et al., 2019). Since it was proposed, many efforts have attempted to improve upon it, and common strategies for doing so are to use more data and train longer (Liu et al., 2019), or to pre-train from scratch on domain-specific data (Gu et al., 2020).

Bidirectional Encoder Representations from Transformers (BERT): Although BERT outperformed on eleven NLP tasks, its pre-training phrase can be optimized further. It is pre-trained on two unsupervised tasks one of which is masked language model (MLM). It trains the model to utilize bidirectionality by predicting a few masked tokens in a sentence. The masking strategy creates a mismatch in the fine-tuning tasks as [MASK] tokens never occur in fine-tuning data until created externally. To soften this, the [MASK] token was used only 80% of the time. Also, in sentences with more than one masked token, the model treats them independent of each other, i.e., the second masked token is assumed to be semantically independent of the first masked token. This assumption hurts the prediction in cases of high dependence between the tokens. The second pre-training objective, next sentence prediction (NSP), improves the performance on Natural Language Interface and Question Answering tasks.

XLNet [11] Inspired by recent successes in bidirectional context representation and segment recurrence, XLNet integrates both into a generalized autoregressive pre-trained model. Using the permutation LM pretraining objective, samples of different orders of the factorization are obtained using different attention masks. This permutation allowed such orders that required contexts of both the left and the right tokens for predicting a token, equipping a bidirectional

behavior. To facilitate such behavior, XLNet’s Two-Stream self-attention architecture produces two kinds of token representations instead of one, i.e., content and query representation. The content representation is the same as in the vanilla transformer and is used when predicting other tokens. On the other hand, the novel query representation is used when the token itself is being predicted. It includes contextual information of tokens occurring before the current token in the order and the positional information of the token itself.

RoBERTa [49] Although self-training models such as XLNet and BERT brought significant performance gains, it is difficult to determine which aspects of the training contributed the most, especially as training is computationally expensive and only limited tuning can be done. A replication of the BERT study that emphasized the effects of training set size and hyperparameter tuning found BERT to be undertrained, and thus RoBERTa is an improved method for training BERT in order to increase performance. Modifications made include training the model longer on larger data sets, changing the masking pattern applied to training data, training on longer sequences, and removing next-sentence prediction. RoBERTa uses the same BERT optimization hyperparameters, with the exception of the peak learning rate and number of warmup steps, which are tuned independently for each setting. The model has proven highly effective, establishing a new state-of-the-art on 4/9 GLUE tasks (MNLI, QNLI, RTE, and STS-B) and matching state-of-the-art results on the RACE and SQuAD datasets.

Text-To-Text Transfer Transformer (T5): T5 combines the results of a rigorous study done across different aspects of a model. It is a deep bidirectional pre-trained model that uses both the encoder and decoder model of the original transformer. The model is pre-trained on Colossal Clean Crawled Corpus (C4), a large dataset created to avoid repetitions, as models underperform when trained on a limited size dataset due to memorization of the repeated examples. A predefined prefix is added before every input sequence to perform a certain downstream task. Although STS-B [24] is a regression task (outputs a number between 1-5), T5 successfully transforms it into a 21-class classification problem. The output is rounded and assigned a class name, thereby fitting a regression task into the framework. Currently, T5 tops the STS-B leaderboard with T5-11B achieving new state-of-the-art solution.

BART [45] is a denoising autoencoder used for pretraining sequence-to-sequence models. BART is particularly effective when fine-tuned for text generation but also works well for comprehension tasks. BART uses a standard Transformer based neural machine translation architecture, which can be seen as generalizing BERT (due to the bidirectional encoder), GPT (with the left-to-right decoder), and many other pretraining plans. BART was pretrained by using corrupted documents then optimizing a reconstruction loss. If all information was lost for source, BART would be equivalent to a language model. Although BART does perform similar to the RoBERTa model, it brings new state-of-the-art results on number of text generation.

Chapter 4

Neural Network Models for Predicate Classification

In this chapter, we introduce methods of fine-tuning BERT for Predicate Classification and our solution to overcome data imbalance problem.

4 Neural Network Models for Predicate Classification

4.1 Problem Setting

Our goal is to identify predicates triggered in a user utterance. For instance, we would like to determine the predicate “*like_activity*” in the utterance “*I like reading book*”. In this thesis, we used a predicate set of 60 predicates listed in Table 1, which was used in Dialogue NLI dataset [3].

No	Predicate	No	Predicate	No	Predicate
1	<i>attend_school</i>	21	<i>has_ability</i>	41	<i>like_read</i>
2	<i>dislike</i>	22	<i>has_age</i>	42	<i>like_sports</i>
3	<i>employed_by_company</i>	23	<i>has_degree</i>	43	<i>like_watching</i>
4	<i>employed_by_general</i>	24	<i>has_hobby</i>	44	<i>live_in_citystatecountry</i>
5	<i>favorite</i>	25	<i>has_profession</i>	45	<i>live_in_general</i>
6	<i>favorite_activity</i>	26	<i>have</i>	46	<i>marital_status</i>
7	<i>favorite_animal</i>	27	<i>have_chidren</i>	47	<i>member_of</i>
8	<i>favorite_book</i>	28	<i>have_family</i>	48	<i>misc_attribute</i>
9	<i>favorite_color</i>	29	<i>have_pet</i>	49	<i>nationality</i>
10	<i>favorite_drink</i>	30	<i>have_sibling</i>	50	<i>not_have</i>
11	<i>favorite_food</i>	31	<i>have_vehicle</i>	51	<i>other</i>
12	<i>favorite_hobby</i>	32	<i>job_status</i>	52	<i>own</i>
13	<i>favorite_movie</i>	33	<i>like_activity</i>	53	<i>physical_attribute</i>
14	<i>favorite_music</i>	34	<i>like_animal</i>	54	<i>place_origin</i>
15	<i>favorite_music_artist</i>	35	<i>like_drink</i>	55	<i>previous_profession</i>
16	<i>favorite_place</i>	36	<i>like_food</i>	56	<i>school_status</i>
17	<i>favorite_season</i>	37	<i>like_general</i>	57	<i>teach</i>
18	<i>favorite_show</i>	38	<i>like_goto</i>	58	<i>want</i>
19	<i>favorite_sport</i>	39	<i>like_movie</i>	59	<i>want_do</i>
20	<i>gender</i>	40	<i>like_music</i>	60	<i>want_job</i>

Table 1 Predicate Categories

In this thesis, we leverage binary relevance method [30]. Binary relevance transforms a multi-label classification problem with L labels into L separate single-label binary classification problems. In this approach, each classifier predicts the membership of one class. The union of the predictions of all the classifiers is taken as the multi-label output.

4.2 BERT-based Models for Predicate Classification

In this study, we used a pre-trained BERT model and fine-tune on our data to detect multiple predicates triggered from the given user utterance for predicate classification. As it is illustrated in Figure 3, we will be creating a neural network with the **Predicate Classifier**. We make use of the BERT Base model which outputs a vector of length 768 for each word (token) and also for the pooled output (CLS). On the top of BERT base model, we add a *Dropout* and *Linear* Layer. They are added for the purpose of *Regularization* and *Classification* respectively. In the forward loop, there are 2 outputs from the BertModel layer. The second output or called the pooled output is passed to the *Dropout* layer and the subsequent output is given to the *Linear* layer. The pooled output at the end of the model training cycle gathers sufficient context of the task and is able to help in making predictions. Note that the number of dimensions for *Linear* Layer is 60 because that is the total number of predicates in which we are looking to classify our model. The primary change here is the usage of Binary cross-entropy with logits (*BCEWithLogitsLoss*) loss function instead of vanilla cross-entropy loss (*CrossEntropyLoss*) that is used for multiclass classification. Binary cross-entropy loss allows our model to assign independent probabilities to the labels.

The model's output are the probabilities of all classes, we hence set a threshold value to pick up a list of most relevant classes to the input utterance as final output.

$$\bar{Y} = \sigma(W_i * X_i + b_i)$$

where X_i referred to input utterance i and σ donates the *sigmoid* function and b bias respectively.

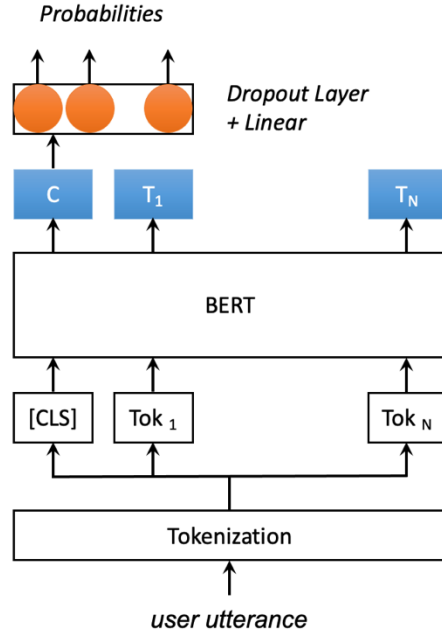


Figure 2: Overview of the BERT model for multi-predicates classification

In this thesis, we leverage Conversational BERT, a pretrained language model with transformer architecture. Conversational BERT (English, cased, 12-layer, 768-hidden, 12-heads, 110M parameters) was trained on the English part of Twitter, Reddit, DailyDialogues [26], OpenSubtitles [27], Debates [28], Blogs [29], Facebook News Comments. We used this training data to build the vocabulary of English sub tokens and took English cased version of BERT-base as an initialization for English Conversational BERT.

4.3 Enriching Training Data from External Datasets

Figure 4 represents the sample distribution per predicate category of the training set. When labels are imbalanced, standard classifiers are usually biased towards the majority label. In this case, a shift is necessary from the general paradigm that optimizes the overall classification accuracy to one that emphasizes the trade-off between precision and recall. In order to solve the problem of data imbalance we use various methods and techniques, and enriching data is one of those.

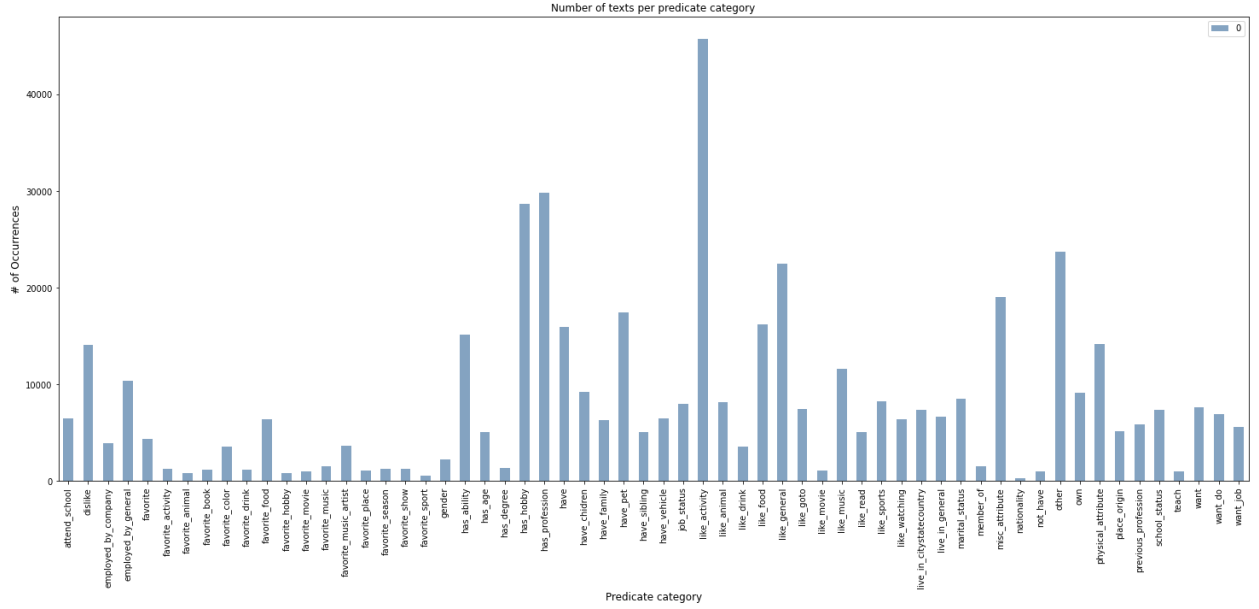


Figure 3: Predicate distribution of the training set

By this approach, we collect more data for labels which have very few samples or no sample such as *favorite_animal*, *favorite_sport*, *nationality*, ...etc by adding more samples from the Multiple-triples SPO dataset and the Dialogue NLI Extra [3] dataset. We briefly introduce these datasets.

Multiple-triples SPO: The Multiple-triples SPO dataset is a human-annotated dataset for user attributes extraction. We collected data from natural conversation, human definition, and from the internet. We manually labelled it with the list of predefined personal attributes. As in the task definition, the mainly purpose of building this dataset is to build a software system which is capable of extracting multiple SAO (*Subject - Attribute - Object*) triples from a single text sentence. The Attribute is referred to a property of the Subject, for example “*age*”, while the Object is its value in context, for example “*40*”. The subject should not be fixed as “*I*” but conform to the real subject in the context. The Attribute is in a list of predefined personal attributes, which currently contains 60 different attributes, and each data set contains about 50 sentences for each attribute. The list of predefined personal attributes such as location (*address*, *hometown*, *birth_place*, *work_location*), organization (*school*, *department*, *company*) ...etc.

Dialogue NLI Extra Dataset [3]: The Dialogue NLI Extra dataset is an extra set of the Dialogue NLI, and it is not overlap the Dialogue NLI dataset. The extra dataset contains additional

(utterance, utterance) pairs. The Dialogue NLI dataset created by Welleck et al. at 2019, it used to improve the consistency of a dialogue model. Researchers noted, “*Dialogue generation can be framed as next utterance prediction.*” The model received input in the form (premise; hypothesis), as well as an associated label generated via human intelligence. Each pair was marked according to these rules: the premise entails the hypothesis, the premise is neutral with respect to the hypothesis, or the premise contradicts the hypothesis. The pairs were thus labeled as “*contradiction,*” “*neutral,*” or “*entails.*”. The extra dataset is containing training set (2,770,000 pairs), validation set (249,000 pairs), test set (237,000 pairs) in JSON file format.

4.4 Using class weight for imbalanced dataset

Most machine learning algorithms are not very robust for biased class data. However, we can modify the existing training algorithm to consider the skew distribution of classes. This can be achieved by giving different weights to the majority and the minority. One of the easiest ways to counter class imbalance is to use class weights wherein we give different weightage to different classes. The number of samples in the classes is considered while computing the class weights. We apply more significant weight to a minority class, which places more emphasis on that class. The classifier thus learns equally from both the classes.

Class weights regularize the loss function. By misclassifying the minority class, a higher loss is incurred by the model since the minority class has a higher weight. This forces the model to learn representations for the minority class. This, however, comes at a price of slightly reduced performance for the majority class. The overall purpose is to punish the misclassification of a few classes by setting higher class weights and reducing the weights for most classes. In the process of training, we give a higher weight to the minority class in the cost function of the algorithm, so that it can provide higher punishment to the minority class, so that the algorithm can focus on reducing the error of the minority class.

Chapter 5

Neural Network Models for Entity Generator

In this chapter, we present sequence to sequence model and how to implement sequence to sequence for Entity Generator.

5 Neural Network Models for Entity Generator

5.1 Entity Generator as Seq2Seq Problem

In order to extract subject (S) and object (O) from generated predicate (P) and user utterance, we consider entity generator as a sequence-to-sequence problem which refers to the problems where either the input and/or the output is a sequence of data (*words, letters...etc.*). In this approach, we leverage sequence-to-sequence (Seq2Seq) [31] which have been designed for sequence learning. Basically, a Seq2Seq model consists of two recurrent neural networks (RNN) as its encoder and decoder, respectively, through which the model cannot only deal with inputs and outputs with variable lengths separately, but also be trained end-to-end. Seq2Seq models can use different settings for the encoder and decoder networks, such as the number of input/output units, ways of stacking layers, dictionary, etc. We generate an input sequence “*summarize: have_chidren ; <s> i have 3 kids </s>*” from generated predicated “*have_chidren*” and original user utterance “*i have 3 kids*” with corresponding output sequence “*i ; 3 kids*”. Now, we consider a dataset of input-output pairs (X, Y) , where $X = (x_1, x_2, \dots, x_{|X|})$ and $Y = (y_1, y_2, \dots, y_{|Y|})$ are the input and output sequences, respectively.

During training, the goal is to learn the relationships between X and Y , which can be formulated as maximizing the Seq2Seq model probability of Y given X :

$$\max p(Y|X) = \max \prod_{t=1}^{|Y|} p(y_t | y_{<t}, X)$$

where $y_{<t} = (y_1, y_2, \dots, y_{t-1})$ are the ground truth tokens of previous steps.

If there are multiple predicates classified, we decode multiple times using the same user utterance for the entity generator with different predicates as input. In this way, we expect our model to transfer knowledge between different predicate generations.

5.2 Seq2Seq Modeling

Sequence-to-sequence (Seq2Seq) models [31] have been designed for sequence learning. Generally, a Seq2Seq model consists of two recurrent neural networks (RNN) as its encoder and decoder, respectively, through which the model cannot only deal with inputs and outputs with variable lengths separately, but also be trained end-to-end.

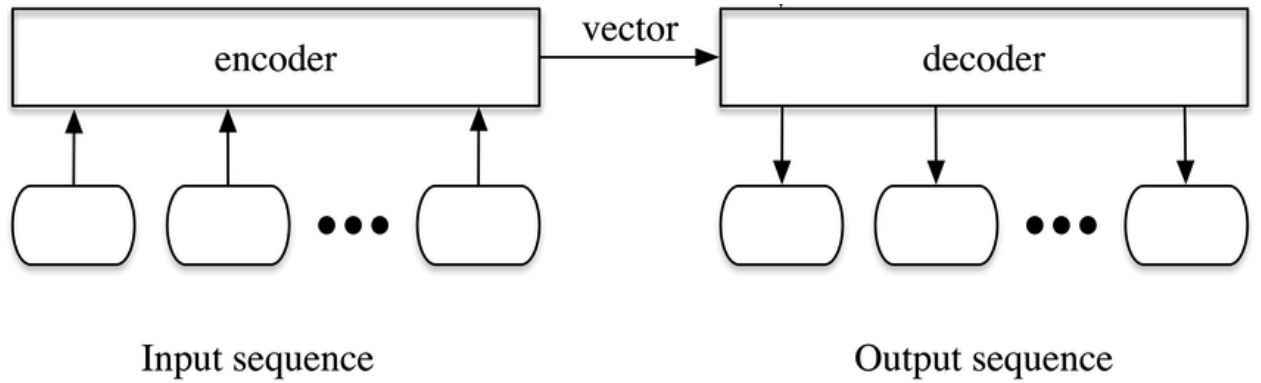


Figure 4: Overview of Sequence-to-Sequence Architecture

The RNNs in the encoder and decoder can be simple RNNs, LSTMs, or GRUs. In a simple RNN, every hidden state is computed using the equation:

$$H_t(\text{encoder}) = \phi(W_{HH} * H_{t-1} + W_{HX} * X_t)$$

Where ϕ is the activation function, $H_t(\text{encoder})$ represents the hidden states in an encoder, W_{HH} is the weight matrix connecting the hidden states, and W_{HX} is the weight matrix connecting the input and the hidden states. The hidden states in the decoder can be computed as follows:

$$H_t(\text{decoder}) = \phi(W_{HH} * H_{t-1})$$

The output generated by the decoder is given as follows:

$$Y_t = H_t(\text{decoder}) * W_{HY}$$

Where W_{HY} is the weights, matrix connecting the hidden states with the decoder output.

After showing promising results in machine translation (MT) tasks [31, 32], Seq2Seq models also proved to be effective for tasks like question answering [33], dialogue response generation [34], text summarization [35], constituency parsing [36], image captioning [37], and so on. Seq2Seq models form the cornerstone of modern response generation models [34, 38, 39, 40].

5.3 Using T5 for Entity Generator

T5 is a new transformer model from Google that is trained in an end-to-end manner with text as input and modified text as output. T5 stands for Text-to-Text Transfer Transformer. It was released by Colin Raffel et al (Oct 2019) [4]. As the name suggested, it's a transformer-based language model used for text generation. In contrast to other famous Transformer based models like BERT or GPT, which were made up of either the encoder or decoder part of the Transformer, T5 paper showed that using the complete encoder-decoder architecture is better than only using decoder. Apart from this, the paper also curated and released Colossal Clean Crawled Corpus (C4) - a huge crawled and cleaned dataset for pre-training language model using self-supervised learning (Raffel et al., 2020) [43]. T5 was trained with multi-task learning methodology, where the idea is to club multiple tasks while pre-training the model 1) Unsupervised training: this includes training on the C4 dataset using the classic language model training with maximum likelihood objective. 2) Supervised training: this includes adding several NLP based tasks like question-answering, summarization, classification, etc. The model is trained with the curated training data in a supervised fashion, but all these tasks are transformed to work with text-in text-out format as suitable for language models. T5 achieves state-of-the-art results on multiple NLP tasks like summarization, question-answering, machine translation, text generation...etc using a text-to-text transformer trained on a large text corpus.

In this study, we consider the entity generation as sequence-to-sequence task. We make use pre-trained T5 model for Entity Generation. If a predicate is triggered, we generate an input sequence with prefix "*summarize:*" from generated predicated and original user utterance then our entity

generation will generate the corresponding subject and object phrases to complete the final user attribute.

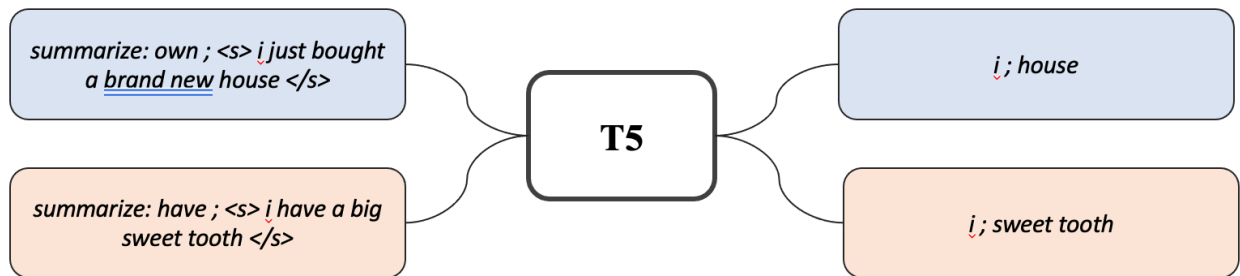


Figure 5: Overview of pre-trained T5 Sequence-to-Sequence model for entity generator task

As it is illustrated in Figure 6, both the subject and object can have more than one word, and we manually concatenate them into one sequence separated by a semicolon. The entity generator will decode multiple times for every classified predicate.

Chapter 6

Experiments and Results

In this chapter, we conducted intensive experiments to investigate different fine-tuning methods of BERT on multi-predicates classification task and T5 on sequence-to-sequence task. Finally, the proposed solution obtained new state-of-the-art results on dataset for user's attribute extraction.

6 Experiments and Results

6.1 Datasets

In this study, we used the Distant Supervision Data [41]. The Distant Supervision Data leveraged two datasets Persona-Chat and Dialogue NLI. If a user utterance and a persona sentence are positively entailed, then the persona triplet of that persona sentence can be represented as one of the possible user attributes. For example, if the utterance “*I prefer basketball; team sports are fun*” and the persona sentence “*I like playing basketball*” has an entailment relationship, then they assign the triplet of the persona sentence labeled by Dialogue NLI, which is (*I, like sports, basketball*), to be one of the user attributes. The data is split into a training, validation, and test set using an 80-10-10 split. The dataset is containing training set (2,770,000 pairs), validation set (249,000 pairs), test set (237,000 pairs) in JSON file format.

6.2 Evaluation Measures

Precision, Recall and F-score The main evaluation metrics of user attributes extraction are precision, recall and F-score. *Precision* (P) is the ratio of SPO triplets correctly identified by the model over all triplets extracted by the model. *Recall* (R) is the ratio of SPO triplets correctly identified by the model over SPO triplets in ground-truth data. *F-score* is the harmonic mean of precision and recall. The *F-score* is calculated by the following equation:

$$F = \frac{2 \cdot P \cdot R}{P + R}$$

Below is a more complex formula for measuring the F-score:

$$F = \frac{(\beta^2 + 1) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

where β is a weighting factor that favours precision when $\beta > 1$ and favours recall when $\beta < 1$.

F_1 score (macro-averaged) The macro-averaged F_1 score is a familiar extension of the F_1 score for binary classification. To compute this metric, however, we average the F_1 scores of all label columns in the data. Similar to traditional F_1 scores, the macro- F_1 score for each label is simply the harmonic mean between precision, p_i , and recall, r_i , for the i^{th} label column:

$$F_{1_i} = \frac{2 \times p_i \times r_i}{p_i + r_i}$$

We then calculate the macro F_1 score by averaging the scores for each label. We denote this as

$$\text{Macro } F_1 = \frac{1}{L} \sum_{i=1}^{|L|} F_{1_i}$$

where L is the total number of label columns.

F_1 score (micro-averaged) In contrast to Macro- F_1 , Micro- F_1 computes statistics globally for all labels and instances. We express this mathematically as

$$\text{Micro } F_1 = \frac{\sum_{n=1}^{|N|} \sum_{i=1}^{|L|} \bar{y}_i^n \times y_i^n}{\sum_{n=1}^{|N|} \sum_{i=1}^{|L|} \bar{y}_i^n + \sum_{n=1}^{|N|} \sum_{i=1}^{|L|} y_i^n}$$

where $\bar{y}_i^n$ and y_i^n are the predicted and actual values, respectively, for label i and instance n . As aptly noted by [13], label columns with few examples strongly affect macro- F_1 , whereas labels with many examples exert greater influence on micro- F_1 .

For reference, we also report the accuracy, and F1 score between the attributes of Distant Supervision Data and the generated attributes. Accuracy and F1 score are computed by strict

matching; i.e., the generated attributes are considered as true positive if and only if every token is exactly the same as the expected attributes.

Accuracy is an evaluation metric that allows you to measure the total number of predictions a model gets right. The formula for accuracy is below:

$$Accuracy = \frac{TrueNegatives + TruePositive}{TruePositive + FalsePositive + TrueNegative + FalseNegative}$$

Accuracy will answer the question, what percent of the models predictions were correct? Accuracy looks at True Positives and True Negatives. A confusion matrix displays counts of the True Positives, False Positives, True Negatives, and False Negatives produced by a model.

There are 4 important terms:

- True Positives: The cases in which we predicted YES and the actual output was also YES.
- True Negatives: The cases in which we predicted NO and the actual output was NO.
- False Positives: The cases in which we predicted YES and the actual output was NO.
- False Negatives: The cases in which we predicted NO and the actual output was YES.

Using a confusion matrix we can get the values needed to compute the accuracy of a model.

6.3 Predicate Classification Evaluation

The predicate classifier is trained using the Adam optimizer [44] with batch size of 32. The learning rate 1e-05, and a 0.1 dropout ratio is used. The loss function used will be a combination of Binary Cross Entropy which is implemented as *BCELogits Loss* in PyTorch. In order to train for maximum per-class accuracy, we have to specify class weights that are inversely proportional to the size of the class. As it is illustrated in Table 2, how to set class weights for imbalanced classes for training the predicate classifier.

Predicate	Weight	Predicate	Weight	Predicate	Weight
<i>attend_school</i>	2.26	<i>has_ability</i>	1.3	<i>like_read</i>	2.59
<i>dislike</i>	1.49	<i>has_age</i>	2.35	<i>like_sports</i>	1.81
<i>employed_by_company</i>	2.8	<i>has_degree</i>	3.55	<i>like_watching</i>	2.31
<i>employed_by_general</i>	1.81	<i>has_hobby</i>	1.0	<i>live_in_citystatecountry</i>	2.13
<i>favorite</i>	2.68	<i>has_profession</i>	1.0	<i>live_in_general</i>	2.18
<i>favorite_activity</i>	3.57	<i>have</i>	1.0	<i>marital_status</i>	1.98
<i>favorite_animal</i>	3.45	<i>have_chidren</i>	1.89	<i>member_of</i>	3.26
<i>favorite_book</i>	4.12	<i>have_family</i>	2.02	<i>misc_attribute</i>	1.21
<i>favorite_color</i>	2.83	<i>have_pet</i>	1.25	<i>nationality</i>	5.35
<i>favorite_drink</i>	4.08	<i>have_sibling</i>	2.56	<i>not_have</i>	4.2
<i>favorite_food</i>	2.31	<i>have_vehicle</i>	2.31	<i>other</i>	1.0
<i>favorite_hobby</i>	3.45	<i>job_status</i>	1.17	<i>own</i>	1.93
<i>favorite_movie</i>	3.35	<i>like_activity</i>	1.0	<i>physical_attribute</i>	1.48
<i>favorite_music</i>	3.63	<i>like_animal</i>	2.08	<i>place_origin</i>	2.52
<i>favorite_music_artist</i>	1.97	<i>like_drink</i>	2.74	<i>previous_profession</i>	2.34
<i>favorite_place</i>	3.87	<i>like_food</i>	1.0	<i>school_status</i>	2.15
<i>favorite_season</i>	3.54	<i>like_general</i>	1.0	<i>teach</i>	3.8
<i>favorite_show</i>	4.01	<i>like_goto</i>	2.08	<i>want</i>	2.16
<i>favorite_sport</i>	4.8	<i>like_movie</i>	3.14	<i>want_do</i>	1.64
<i>gender</i>	3.34	<i>like_music</i>	1.69	<i>want_job</i>	2.44

Table 2 Class weights definition for imbalanced classes

6.3.1 Baselines

In the user attributes extraction from dialogue area to the best of our knowledge, no previous work has explicitly modeled the consistency issue. To evaluate our model, we compared the proposed approach with the following strong deep learning neural networks by using the same model, but replace pretrained Conversational BERT by other pretrained language models:

End-to-End Memory Network [46] is a neural network with a recurrent attention model over a possibly large external memory. The architecture is a form of Memory Network, but unlike the model in that work, it is trained end-to-end, and hence requires significantly less supervision during training. It can also be seen as an extension of RNN search to the case where multiple computational steps (hops) are performed per output symbol.

SqueezeBERT [47] is an efficient architectural variant of BERT for natural language processing that uses grouped convolutions. It is much like BERT-base, but with positional feedforward connection layers implemented as convolutions, and grouped convolution for many of the layers.

XLNet [11] a generalized autoregressive pretraining method that (1) enables learning bidirectional contexts by maximizing the expected likelihood over all permutations of the factorization order and (2) overcomes the limitations of BERT thanks to its autoregressive formulation.

DistilBERT [48] learns a distilled (approximate) version of BERT, retaining 97% performance but using only half the number of parameters. Specifically, it does not have token-type embeddings, pooler and retains only half of the layers from Google’s BERT. DistilBERT uses a technique called distillation, which approximates the Google’s BERT, i.e., the large neural network by a smaller one.

Table 3 lists the performance results of the baseline and proposed methods in predicate classification from user utterances.

	Accuracy	F1-Score (micro-averaged)
<i>End-to-End Memory Network (baseline)</i>	41.57	44.40
<i>SqueezeBERT</i>	52.80	52.80
<i>XLNet</i>	43.91	51.24
<i>DistilBERT</i>	35.84	45.55
<i>Ours (Without Data Enrichment)</i>	54.50	61.79
<i>Ours</i>	73.25	79.04

Table 3 Comparison of multi-predicate classification for baselines and proposed methods

6.3.2 Results

Table 3 gives the accuracy of the baseline and proposed methods in predicate classification. A comparison between the baseline and proposed methods indicates that the proposed method using pre-trained BERT improved the accuracy by 31.68%. The experimental results showed the proposed method’s effectiveness. With the proposed method, the accuracy was greatly improved to 73.25% from 41.57% in the predicate classification.

6.4 Entity Generator Evaluation

We used T5 to conduct the entity generator, a pre-trained and very large (e.g., roughly twice the size of BERT-base) encoder-decoder Transformer model. Model are fine-tuned on our Distant Supervision Dataset before being evaluated. In this study, we will take a pretrained T5-base model and fine tune it to generate a one-line summary of source text using PyTorch with a batch size of 24 for both train and val. The guide proceeds by (1) preparing the data for sequence-to-sequence model with T5 small – a small version of T5 base, and (2) training the data in PyTorch.

Data preparation: Some unique pre-processing is required when using T5 for sequence-to-sequence model. Specifically, we need to add prefix “*summarize:*” to the beginning of all of our text that needs to be summarized and we need shift our summaries rightward by one token which we do by adding triggered predicate to the beginning of each utterance. Each utterance will be surrounded by a pair of tags (*<s></s>*) and separated from the triggered predicate by a semicolon (;), for example “*summarize: own ; <s> i just bought a brand new house </s>*”. T5’s tokenizer in the transformers library will handle the details from there. The corresponding label will be “*i ; house*”, both the subject (*S*) and object (*O*) can have more than one word, and separated by a semicolon (;).

Training model: We want to look at the average length of the text to decide if input can be tokenized to a max length of 256. For Distant Supervision Dataset, the average length of text is 192 words and average length of output text is 78. In this practice, I have used an input target length of 256 tokens and an output summary length of 100. The model was trained for 2 epochs and WANDB logger showed good improvement in ROUGE1 score and Val loss as the model trained.

6.4.1 Baselines

For experimental setup, we compare the entity generation with state-of-the-art supervised approaches for text-to-text and evaluated on our dataset. We briefly introduce the baselines:

Attention Is All You Need [19], a great advance in the use of the attention mechanism, being the main improvement for a model called Transformer. Attention mechanisms are one of the

solutions to overcome the problem of model forgetting. This is because they allow dependency modelling without considering their distance in the input or output sequences.

BART is a denoising autoencoder for pretraining sequence-to-sequence models, which stands for Bidirectional and Auto-Regressive Transformers [45]. It essentially generalizes BERT and GPT based architectures by using the standard Seq2Seq Transformer architecture from [19] while mimicking BERT/GPT functionality and training objectives. As described in their paper, BART is trained by (1) corrupting text with an arbitrary noising function, and (2) learning a model to reconstruct the original text. As a result, BART performs well on multiple tasks like abstractive dialogue, question answering and summarization. Specifically, for summarization, with gains of up to 59.86%.

6.4.2 Results

	Accuracy	F1-Score
<i>User Attribute Extraction from Dialogues (baseline)</i>	43.48	46.03
<i>Attention Is All You Need</i>	41.53	41.09
<i>BART</i>	61.78	59.86
<i>Ours (Without Data Enrichment)</i>	62.61	62.61
<i>Ours</i>	62.61	64.29

Table 4 Comparison of entity generation for baselines and proposed methods

The results of our experiments are summarized in Table 4, the proposed entity generation model achieves the highest F1 score, 64.29%. A comparison between the baseline and proposed methods indicates that the F measure of the proposed method improved by 18.26%.

6.5 SPO Evaluation

In this experiment, we evaluate the proposed attribute extractor’s performance, which has a predicate classifier, and an entity generator in unseen setting (i.e., all labels are unseen during training). We present attribute-extraction experimental results for 60 predicates. If there are multiple predicates classified, the generator will decode multiple times for every triggered predicate. The baseline method used was that of [41]. We use accuracy and F measures to

evaluate the proposed user attributes extraction performance. The results of the seen experiment are presented in Table 5.

	Accuracy	F1-Score
<i>baseline</i>	26.52	28.68
<i>BERT+BART</i>	40.61	41.96
<i>Ours (Without Data Enrichment)</i>	41.28	42.41
<i>Ours</i>	49.74	50.85

Table 5 Comparison of user’s attribute extractor for baselines and proposed methods

A comparison between the baseline and proposed method indicates that the F measure of the proposed method improved by 22.17%. With the proposed method, the accuracy was greatly improved to 28.68% from 50.85% in the *unseen* test dataset. We also reported the result of proposed method on the original Distant Supervision Dataset before enriched, the proposed method obtained 42.41%. This indicates that proposed method outperforms baseline methods, and data enrichment has significantly improved the model’s performance.

Our error analysis indicated that 40% of errors consisted of a failure to include analyzing context of user utterances. For examples, from the pair “*i drive a ford pickup truck.*” – “*sure, i pick him up for church every Sunday with my ford pickup*” the system would not extract the correct $\langle\langle i, have_vehicle, ford \rangle\rangle$ because of out-of-context. This indicates that context in the user utterance does severely damage the proposed method’s performance. We find three common errors: wrong predicate prediction, ambiguous attribute inference, and missing attribute prediction. First, if our model does not predict predicates correctly, it may generate out-of-context object phrases. For example, our model predicts like music as a triggered predicate for the utterance “*I like classic cars!*” because it is biased by people mentioning classical music. Second, we find that in some cases our model generates attributes that are relevant but not certain, making the attribute ambiguous. For example, when a user says he/she is “*Tired from too many parties,*” our model predicts the attribute (*I, like_activity, partying*) although the user does not mention it explicitly. Third, sometimes no predicate is triggered, even if there is some useful user information. For example, we should be able to conclude that a user likes to travel if he/she says “*I travel a lot. I even studied abroad.*”

Chapter 7 Conclusion

7 Conclusion

Extracting user's attribute from conversations is a crucial step towards personalizing chatbots, search and recommendations. Conversational data contains a wealth of such knowledge, which is, however, challenging to extract. The previous research attempts have mostly been limited to extraction from explicit mentions or classification of trivial attributes. Therefore, I find it a topical and intriguing issue to work upon. In this report, I presented our current research for inferring user attributes from conversations. Our work differs from prior work by its ability to predict multiple attribute values from user utterance. We demonstrated the ability of our model for inferring users' attribute values through comprehensive experiments with unseen conversations. These models show promising results, quantitatively outperformed the baselines, and gave qualitatively interpretable predictions. As shown in Table 5, the proposed attribute extraction model achieves the highest F1 score, 50.85%, which surpassed the other two generation models. The predicate classifier achieves a 79.04% F1 score on the multi-label classification with 60 possible predicates. In the oracle study, the entity generator, which is given the correct predicates in the Distant Supervision Data as input, can obtain a 64.29% F1 score. For instance, “*yes. two dogs and a cat. they are my babies.*” our model is able to predict (*i, have_pet, 2 dog*). We have presented the idea of extracting user attributes from daily dialogues. Although our model with Distant Supervision Data can achieve reasonable results, we believe there exist limitations that should be addressed in the future. We also provided some negative examples of our generated user attributes. We found three common errors: wrong predicate prediction, ambiguous attribute inference, and missing attribute prediction. For example, our model predicts *like_sports* as a triggered predicate for the utterance “*my favorite sport is ultimate frisbee.*” because it is biased, or our model predicts *favorite_food* as a triggered predicate for the user utterance “*i would rather eat chocolate cake during this season.*”.

Once we obtained user attributes, they can be applied to many downstream applications, for example, search, friend recommendation, online advertisement, computational social science, personalized personal assistant, etc. In terms of future work, we see many areas of improvement

and expansion. One idea would be increasing the size of the dataset by using the best classifier, the one trained with BERT, to label samples from the unlabeled dataset (or from newly crawled documents) and manually check them afterwards. It would be interesting to see how much the results might improve by, for instance, doubling the size of the labelled dataset. Additionally, we plan to implement a personalized dialogue system using extracted user information. We also want to solve the problems by applying the context to extract more user information and to apply our method to other languages.

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